SOFTWARE REQUIREMENTS SPECIFICATION

Simulink Demo Application — SCU / BCM CAN Communication

Based on STLA_Whitepaper.drawio

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1. Purpose

This document defines the requirements for a simple Simulink demonstration model that replicates the CAN communication behaviour shown in STLA_Whitepaper.drawio. The model shall showcase two virtual ECU nodes — SCU and BCM — exchanging three types of CAN frames: a startup pre-condition frame, a periodic position frame, and an on-demand request-response pair. The goal is a runnable, visual demo that any engineer can follow without prior knowledge of the VECU platform.

2. What the Drawio Shows

The diagram defines two nodes and three distinct communication flows. The table below maps every labelled element in the drawio to its role in the Simulink model.

Drawio Element	Type	Meaning
SCU — app.seatApp.elf	Node	The seat controller. Sends position data. Responds to BCM requests.
BCM — app.reference.elf	Node	The body controller. Sends the pre-condition frame and on-demand requests.
0x101 [00 00 00 00 00 00 00 00]	Pre-condition Frame	Sent once by BCM at startup. SCU must receive it before doing anything. SCU must NOT generate or modify this frame.
Frame_1 (D0) — 100 ms periodic	Periodic TX	SCU transmits seat data to BCM every 100 ms. Starts at t = 500 ms. One-directional: SCU → BCM only.
Frame_2 F1 — On-demand request	Request TX	BCM sends a request to SCU. Starts at t = 600 ms. Event-driven (on-demand).
Frame_2 F2 — Response	Response TX	SCU replies to each F1 request. Bi-directional exchange: BCM → SCU (F1) then SCU → BCM (F2).
Visual Board — vECU_1 / vECU_2	Timing chart	Shows simulation time vs simulation cycle for each node — i.e. when each TX event occurred. This is the visual output of the demo.
Future note on Frame_2	Design note	The drawio marks that the F2 response should eventually become periodic rather than event-driven. Not in scope for this demo.

3. System Overview

The Simulink model shall contain two top-level subsystems representing the two nodes. They communicate by passing signal buses that simulate CAN frames. A shared Clock block provides the simulation time reference. A Visual Board scope replicates the timing chart shown in the drawio.

Node	Simulink Block	Sends	Receives

SCU	Subsystem: SCU	Frame_1 every 100 ms (from t = 500 ms) Frame_2 F2 on each F1 receipt	Pre-condition frame (0x101) at startup Frame_2 F1 on-demand from BCM
BCM	Subsystem: BCM	Pre-condition frame (0x101) once at t = 0 Frame_2 F1 on-demand from t = 600 ms	Frame_1 every 100 ms Frame_2 F2 in response to each F1

NOTE: The model does not need to simulate real seat motor movement or sensor data. The frame payloads can carry simple counter or constant values. The focus is entirely on the communication timing and frame flow shown in the drawio.

4. Requirements

4.1 Pre-Condition Frame (0x101)

Req ID	Requirement	Rationale
PRE-01	The BCM subsystem shall transmit the pre-condition frame (CAN ID 0x101, 8-byte payload of all zeros) exactly once at simulation time t = 0 ms.	The drawio shows this as the first event in the sequence — BCM initiates the session.
PRE-02	The SCU subsystem shall include a one-bit internal flag (Ready_Flag) that is set to 1 upon receipt of the pre-condition frame. The SCU shall not transmit Frame_1 or respond to Frame_2 F1 until Ready_Flag = 1.	Models the startup handshake — SCU must not act before BCM signals readiness.
PRE-03	The SCU subsystem shall not generate, retransmit, or modify the pre-condition frame under any circumstances.	The drawio explicitly marks this frame as 'Not to Touch'. This is a hard constraint.

4.2 Frame_1 — Periodic SCU Transmission (D0)

Req ID	Requirement	Rationale
F1-01	The SCU subsystem shall begin transmitting Frame_1 at simulation time t = 500 ms and repeat every 100 ms thereafter for the duration of the simulation.	The drawio labels this 'at 500 ms' with 'Periodicity x100 ms'. Both values are explicit constraints.
F1-02	Frame_1 shall carry a simple incrementing counter (uint8, wraps at 255) as its payload (D0 byte). All remaining payload bytes shall be zero.	A counter provides visible proof of transmission sequence without requiring real sensor data.
F1-03	Frame_1 shall be transmitted in one direction only — SCU to BCM. The BCM subsystem shall receive and log it but shall not retransmit it.	The drawio annotation explicitly states 'one-directional'.
F1-04	The BCM subsystem shall log the simulation timestamp of each received Frame_1 and output it as a signal to the Visual Board.	Drives the vECU_2 timing chart on the Visual Board, replicating the drawio diagram.

4.3 Frame_2 — On-Demand Request / Response (F1 / F2)

Req ID	Requirement	Rationale
F2-01	The BCM subsystem shall send Frame_2 F1 (request) on-demand starting at t = 600 ms. For the demo, the request shall	The drawio marks 'at 600 ms......' indicating on-

	be triggered by a Manual Switch block so the presenter can send it at any time during the simulation.	demand behaviour starting from this point. A Manual Switch makes it interactive for the demo.
F2-02	Frame_2 F1 shall carry a simple fixed-value payload (e.g. 0xAA in byte 0, all others zero) to act as a recognisable request marker.	Distinguishes F1 from other frames on the bus when viewing the Visual Board.
F2-03	Upon receipt of Frame_2 F1, the SCU subsystem shall generate and transmit Frame_2 F2 (response) within the same simulation step (5 ms).	The drawio shows F2 as a direct response to F1 with no delay — the bi-directional exchange must be immediate.
F2-04	Frame_2 F2 shall echo the F1 request counter in byte 0 and set byte 1 to 0x01 (ACK). All other bytes shall be zero.	The echo proves the SCU received the correct F1 frame. The ACK byte visually confirms the response on the scope.
F2-05	The SCU subsystem shall log the simulation timestamp of each transmitted F2 response and output it as a signal to the Visual Board.	Drives the vECU_1 TX timing plot on the Visual Board.

4.4 Visual Board

NOTE: The Visual Board is the primary demo output. It directly replicates the timing diagram shown in the drawio and is the main thing the audience will see.

Req ID	Requirement	Rationale
VB-01	A Simulink Scope (or Dashboard Scope) named 'Visual Board' shall display two subplots stacked vertically: the top for vECU_1 (SCU) TX events and the bottom for vECU_2 (BCM) TX events. The horizontal axis of both shall be simulation time in seconds.	Directly replicates the two-chart layout shown in the drawio Visual Board section.
VB-02	The SCU (vECU_1) subplot shall display Frame_1 TX timestamps as discrete vertical impulses and Frame_2 F2 TX timestamps as separate discrete impulses on the same axis, distinguishable by amplitude or colour.	Shows when each SCU frame was sent — matching 'Data Points for when TX was done by vECU_1' in the drawio.
VB-03	The BCM (vECU_2) subplot shall display the pre-condition frame TX event (single impulse at t = 0) and Frame_2 F1 TX timestamps as discrete impulses.	Shows when each BCM frame was sent — matching 'Data Points for when TX was done by vECU_2' in the drawio.
VB-04	Both subplots shall share the same simulation time axis range (0 s to end of simulation). The scope shall auto-scale the vertical axis to show all impulse events clearly.	Ensures the timing relationship between SCU and BCM frames is visible at a glance.

4.5 General Model Requirements

Req ID	Requirement	Rationale
GEN-01	The model shall use a fixed-step solver with a step size of 5 ms and a simulation stop time of 2000 ms (2 seconds). This gives enough time to observe at least 15 Frame_1 transmissions and multiple Frame_2 exchanges.	5 ms matches the AUTOSAR Classic OS task period. 2 s provides sufficient demo duration.
GEN-02	The SCU and BCM shall each be implemented as a single Simulink Subsystem block at the top level. Inter-node signals (frames) shall be passed as Bus signals between subsystems.	Clean separation of nodes matches the two-ECU architecture in the drawio.
GEN-03	The model shall require no external toolboxes beyond base Simulink. No Vehicle Network Toolbox, Stateflow, or Embedded Coder licence shall be needed to open or run the model.	Maximises portability — any engineer with a standard Simulink licence can run the demo.
GEN-04	All timing logic (500 ms Frame_1 start, 100 ms periodicity) shall be implemented using a single shared Clock block and Compare To Constant blocks. No hardcoded Delay blocks with magic numbers.	Clock-based timing is transparent and easy to inspect or modify during the demo.
GEN-05	The model shall run in Normal simulation mode and shall not require code generation or compilation to execute.	Keeps the demo simple — press Run, see results.

5. Timing Summary

The table below is the definitive timing reference for the model, derived directly from the drawio annotations.

Frame	Direction	First TX	Periodicity	Trigger
Pre-condition (0x101)	BCM → SCU	t = 0 ms	Once only	Simulation start
Frame_1 (D0)	SCU → BCM	t = 500 ms	Every 100 ms	Timer (cyclic)
Frame_2 F1 (Request)	BCM → SCU	t = 600 ms	On-demand	Manual Switch
Frame_2 F2 (Response)	SCU → BCM	Same step as F1	On-demand	F1 receipt

6. Acceptance Criteria

The demo model passes if all of the following can be observed on the Visual Board scope after a single simulation run — no post-processing, no external tools.

#	Observable Criterion	Verified By
1	A single BCM TX impulse appears at t = 0 ms on the BCM subplot (pre-condition frame).	Visual Board — BCM subplot
2	The first SCU TX impulse for Frame_1 appears at t = 500 ms ± 5 ms on the SCU subplot.	Visual Board — SCU subplot
3	Subsequent Frame_1 impulses are spaced exactly 100 ms apart on the SCU subplot.	Visual Board — SCU subplot
4	Each press of the Manual Switch produces a BCM TX impulse (F1) and a matching SCU TX impulse (F2) within the same 5 ms step on both subplots.	Visual Board — both subplots
5	No SCU TX events appear before t = 500 ms (Ready_Flag gate is working).	Visual Board — SCU subplot
6	Frame_2 F2 byte 0 echoes the F1 byte 0 value. Byte 1 of F2 = 0x01. Verified on a Display block.	Display block on F2 output

7. Out of Scope

The following are explicitly not required for this demo model:

- Real seat motor control or PID position control.
- Actual sensor data — all payloads use counters or constants.
- Fault detection, watchdog timers, or safety monitoring.
- Memory-position save and recall.
- Periodic Frame_2 F2 response (noted as a future enhancement in the drawio but not in scope here).

- AUTOSAR code generation, Embedded Coder, or any toolbox beyond base Simulink.
- Hardware-in-Loop (HIL) or SIL Kit integration.

8. Future Enhancement (from Drawio)

NOTE: The drawio explicitly notes: 'Future: Response has to be periodicity.' Once this demo is validated, the Frame_2 F2 response shall be converted from an event-driven reply to a periodic transmission at the same 100 ms rate as Frame_1. This will align F2 with the one-directional periodic model and simplify BCM receive scheduling.

9. Revision History

Version	Date	Author	Description
1.0	March 2026	KPIT Technologies	Initial release. Requirements scoped exclusively to the communication flows defined in STLA_Whitepaper.drawio: pre-condition frame, Frame_1 periodic, Frame_2 request-response, and Visual Board timing output.